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REPORT

ON

STREAMING CATALOGUE AND RIGHTS (CASE 2C)

CSE 4007 - SEMANTIC WEB

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1. Problem Scope

The aim of the semantic web is to describe data explicitly and formally, so that software can work with its intended meaning instead of matching strings. A streaming catalogue is a good test of this idea: its facts are about things (titles, people, languages, licences) and about how those things are related over time.

1.1 Purpose

A streaming service needs catalogue data that a program can interpret. Titles, people, genres, languages and territories must be resources that other statements can point to, not words typed into a field. Rights information must also be kept apart from the title itself: a licence is a legal right with a territory and a period, and an availability event is the period in which a title is actually offered. When a licence expires the film does not change, only the record about the right does. With this separation, a question such as "which titles can a viewer in India watch today?" is answered by following links and comparing dates.

Let's take an example:- the BBC Programmes Ontology [9] models a programme as Brand, Series and Episode resources, treats a *Version* of an episode as a separate resource, and models a *Broadcast* as an event that refers to a version instead of being a property of the programme. This report follows the same layering: title, language version, licence, availability event.

1.2 In and Out of Scope

Case information	**Modelled as**
Films **Riverline** and **Last Signal**	Two films (`sv:Film`)
Series **City Loop** with episodes **E1–E3**	One series and three episodes (`sv:Series`, `sv:Episode`) with `sv:episodeNumber` 1, 2, 3
Contributors **Dev Rao**, **Maya Sen** and **Jo Paul** in different roles	Three persons (`sv:Person`) Four roles (assumed): Director, Actor, Writer, Narrator Nine credits (`sv:Contribution`), three per person
Genres **Drama** and **Documentary**	Two reusable genres (`sv:Genre`)
English, Hindi and Malayalam versions	Three languages and seven language versions (`sv:LanguageVersion`), each with a runtime
India and Singapore licences with start/end dates	Two territories and three licences (`sv:Licence`), each with `xsd:date` start and end
One expired and two active availability records	Three availability events (`sv:AvailabilityEvent`): a period, a status and an as-of date

Out of scope: pricing, users and subscriptions; ratings and reviews; subtitle and dubbing tracks beyond the seven language versions; credits for individual episodes; licences granted for a series as a whole (rights are recorded per language version); OWL axioms and SHACL constraints.

1.3 Assumptions

The case leaves some details open, so the following are assumed and marked as illustrative in the Turtle file:

- The case says contributors appear “in different roles” but does not name them. Director, Actor, Writer and Narrator are assumed.
- The genre of each title, which language versions exist, runtimes, licence dates and availability dates are sample values chosen to satisfy the case.
- Availability status is a snapshot taken on **2026-10-13**. Start and end dates are inclusive. On that date one availability record has ended and two are running.
- Episodes E1–E3 exist in a Hindi version and are licensed in India. Riverline and Last Signal have two versions each.

2. Semantic Model

Vocabulary IRI:

https://example.org/streaming/vocab#

Prefix	Namespace	Use
`sv:`	`https://example.org/streaming/vocab#`	Classes and properties (the RDFS vocabulary)
`ex:`	`https://example.org/streaming/id/`	All individuals (the catalogue data)
`rdf:`	`http://www.w3.org/1999/02/22-rdf-syntax-ns#`	`rdf:type`, `rdf:Property`
`rdfs:`	`http://www.w3.org/2000/01/rdf-schema#`	`Class`, `subClassOf`, `domain`, `range`, `label`, `comment`, `seeAlso`
`xsd:`	`http://www.w3.org/2001/XMLSchema#`	Datatypes: date, duration, integer

The domain *example.org* is reserved for examples (RFC 2606) [13], so these IRIs cannot clash with a real website.

1) Classes (15)

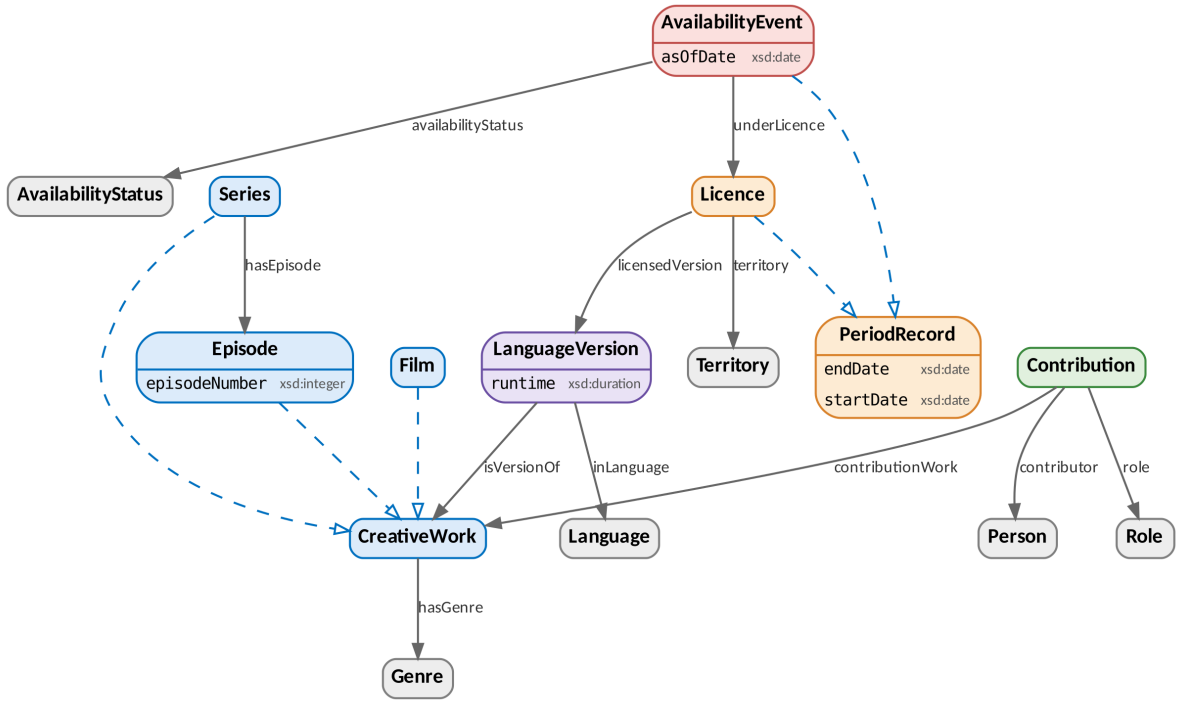
- **CreativeWork**: A title in the catalogue: a film, a series or an episode. It carries no rights information. (`sv:CreativeWork`)
- **Film**: A standalone feature-length creative work. (`sv:Film`)
- **Series**: A creative work that is made up of episodes. (`sv:Series`)
- **Episode**: One instalment of a series. (`sv:Episode`)
- **Person**: A contributor who can be credited on creative works. (`sv:Person`)
- **Role**: A kind of contribution, such as director, actor, writer or narrator. (`sv:Role`)
- **Genre**: A category of creative work, such as Drama or Documentary. (`sv:Genre`)
- **Language**: A language in which a version of a creative work is offered. (`sv:Language`)
- **Territory**: A country or region in which a licence applies. (`sv:Territory`)
- **LanguageVersion**: A creative work as offered in one language. It is the unit that has a runtime and that is licensed. (`sv:LanguageVersion`)
- **Contribution**: A credit that links one person, one role and one creative work. (`sv:Contribution`)
- **PeriodRecord**: A record that holds for a period with a start date and an end date (both inclusive). (`sv:PeriodRecord`)

- **Licence**: The legal right to stream specific language versions in one territory for a rights period. (`sv:Licence`)
- **AvailabilityEvent**: A period during which licensed versions are actually offered on the platform, under one licence. (`sv:AvailabilityEvent`)
- **AvailabilityStatus**: Whether an availability event is active or expired on its as-of date. (`sv:AvailabilityStatus`)

```
rdfs:Resource
|-- sv:CreativeWork
|   |-- sv:Film
|   |-- sv:Series
|   `-- sv:Episode
|-- sv:PeriodRecord
|   |-- sv:Licence
|   `-- sv:AvailabilityEvent
`-- Person, Role, Genre, Language, Territory, LanguageVersion,
    Contribution, AvailabilityStatus   (no subclasses)
```

Class hierarchy

Film, Series and Episode are subclasses of CreativeWork. Licence and AvailabilityEvent are subclasses of PeriodRecord.



Vocabulary diagram (Figure 1)

Solid arrows are object properties from domain to range; dashed arrows are `rdfs:subClassOf`; data properties are listed inside their domain class. Blue: creative works. Green: credits. Purple: language versions. Orange: licences and period records. Red: availability events. Grey: reference data.

2) Object Properties (11)

Property	Domain → Range	Meaning
`hasGenre`	CreativeWork → Genre	Links a creative work to its genre.
`hasEpisode`	Series → Episode	Links a series to one of its episodes.
`isVersionOf`	LanguageVersion → CreativeWork	Links a language version to the creative work it is a version of.

Property	Domain → Range	Meaning
`inLanguage`	LanguageVersion → Language	Links a language version to the language it is offered in.
`contributionWork`	Contribution → CreativeWork	The creative work a contribution (credit) belongs to.
`contributor`	Contribution → Person	The person named in a contribution (credit).
`role`	Contribution → Role	The role the person holds in a contribution (credit).
`licensedVersion`	Licence → LanguageVersion	Links a licence to a language version that it covers.
`territory`	Licence → Territory	The territory in which a licence applies.
`underLicence`	AvailabilityEvent → Licence	Links an availability event to the licence that allows it.
`availabilityStatus`	AvailabilityEvent → AvailabilityStatus	The status (active or expired) of an availability event on its as-of date.

3) Data Properties (5)

Property	Domain → Range	Meaning
`episodeNumber`	Episode → xsd:integer	Position of an episode within its series (1 for E1, 2 for E2, ...).
`runtime`	LanguageVersion → xsd:duration	Running time of a language version, written as an ISO 8601 duration such as PT1H40M.
`startDate`	PeriodRecord → xsd:date	First day of a period record (inclusive).
`endDate`	PeriodRecord → xsd:date	Last day of a period record (inclusive).
`asOfDate`	AvailabilityEvent → xsd:date	The date on which the stated availability status was determined.

Titles and names are not data properties. They are `rdfs:label` values (strings tagged `@en`) on the resource they name, so every relationship in the graph uses an IRI.

4) Instances (44)

Class	No.	Individuals (IRIs under `ex:`)
`sv:Film`	2	Riverline, LastSignal
`sv:Series`	1	CityLoop
`sv:Episode`	3	CityLoop_E1, CityLoop_E2, CityLoop_E3
`sv:Person`	3	DevRao, MayaSen, JoPaul
`sv:Role`	4	Director, Actor, Writer, Narrator
`sv:Genre`	2	Drama, Documentary
`sv:Language`	3	English, Hindi, Malayalam
`sv:Territory`	2	India, Singapore
`sv:LanguageVersion`	7	Riverline_EN, Riverline_HI, LastSignal_EN, LastSignal_ML, CityLoop_E1_HI, CityLoop_E2_HI, CityLoop_E3_HI

Class	No.	Individuals (IRIs under `ex:`)
`sv:Contribution`	9	Credit_Riverline_DevRao_Director, Credit_Riverline_MayaSen_Actor, Credit_Riverline_JoPaul_Writer, Credit_LastSignal_JoPaul_Director, Credit_LastSignal_DevRao_Narrator, Credit_LastSignal_MayaSen_Writer, Credit_CityLoop_MayaSen_Director, Credit_CityLoop_JoPaul_Actor, Credit_CityLoop_DevRao_Writer
`sv:Licence`	3	Licence_IN_Riverline, Licence_SG_LastSignal, Licence_IN_CityLoop
`sv:AvailabilityEvent`	3	Availability_IN_Riverline, Availability_SG_LastSignal, Availability_IN_CityLoop
`sv:AvailabilityStatus`	2	Active, Expired

3. Modelling Decisions

1) Entities as reusable IRIs, not plain text

Choice : Every title, person, role, genre, language, territory, version, credit, licence and availability record is an RDF resource [2] with its own IRI under `ex:` (for example `ex:Riverline`, `ex:MayaSen`, `ex:Hindi`, `ex:India`). Names appear only as `rdfs:label`. IRIs follow one pattern: titles `ex:Riverline`, episodes `ex:CityLoop_E1`, versions `ex:Riverline_HI` (title + language code), credits `ex:Credit_Riverline_DevRao_Director`, licences `ex:Licence_IN_Riverline`. No IRI contains a status or other value that could change. Each language and territory also points to its DBpedia resource with `rdfs:seeAlso` [12].

Alternative : Store them as string literals, for example `ex:Riverline sv:director "Dev Rao"`.

Reasoning : A string cannot be linked or reused. `ex:MayaSen` is the object of three credits, so Q3 can count her three roles on three works. As strings, “Maya Sen” and “Maya Sen” (two spaces) would be two different people and nothing in the graph would say they are the same. Reuse also removes duplication: Hindi is the language of four versions, India is the territory of two licences and Drama is the genre of two titles, yet each exists once. The vocabulary (`sv:`) and the data (`ex:`) have separate namespaces, so another catalogue can reuse the vocabulary without this data.

2) Contribution as a node instead of a fixed role per person

Choice : A role belongs to a credit, not to a person. A `sv:Contribution` links one person, one role and one work through `sv:contributor`, `sv:role` and `sv:contributionWork`. There are nine credits, and each person holds three different roles across the three titles.

Alternative : Give the role to the person (`ex:MayaSen a sv:Actor`) or use one property per role (`sv:directedBy`, `sv:actedIn`).

Reasoning : The case says the contributors appear “in different roles”. Maya Sen is an Actor in Riverline, a Writer in Last Signal and the Director of City Loop (Q2). A role attached to the person would be wrong for two of the three. One property per role would need a new property for every role and still could not carry anything else about the credit. With a credit node, a new role is just a new `ex:Role` individual. The cost is one extra hop in queries.

3) LanguageVersion between a work and its licence

Choice : `sv:LanguageVersion` represents a work as offered in one language (`sv:isVersionOf`, `sv:inLanguage`, `sv:runtime`). Licences list the versions they cover (`sv:licensedVersion`). Rights are therefore recorded per version and per episode, and nothing is inferred from series membership.

Alternative : Link the work directly to a `sv:Language` and attach the licence to the work.

Reasoning : (a) The runtime belongs to the version: Riverline runs PT1H40M in English and PT1H44M in Hindi (Q5). On the work there would be two runtimes and no way to say which is which. (b) Rights are granted per version: the Singapore licence covers the English and Malayalam versions of Last Signal, and the India licence for City Loop covers only the Hindi versions of E1, E2 and E3. (c) The work keeps one identity, with one genre and one set of credits, however many versions exist. The same layering appears in the BBC Programmes Ontology (a Version of an episode) [9] and in FRBR (work and expression) [10].

4) Creative work, licence and availability event as separate layers

Choice : `sv:CreativeWork` holds only the title, genre and credits. `sv:Licence` is the legal right (versions, territory, rights period). `sv:AvailabilityEvent` is the period in which the licensed versions are actually offered, linked to one licence by `sv:underLicence`. None of these classes is a subclass of another, and Q12 confirms that no resource is both a creative work and a version, licence or availability event.

Alternative : Put territory, start date and end date on the work, or use a single “rights” node.

Reasoning : One title can be licensed in several territories and periods, so dates on the work would be ambiguous. A licence is also not the same as availability: the City Loop licence runs from 2026-07-01 to 2027-06-30, but the episodes are offered only from 2026-08-01 to 2027-03-31, and the Last Signal licence runs to 2026-06-30 although the film left the platform on 2026-06-15 (Q8). When that licence ended, nothing about the film changed; only a record did. The case also asks for “one expired and two active availability records”, which are statements about availability, not about the title.

5) One pair of date properties on a shared superclass (PeriodRecord)

Choice : `sv:Licence` and `sv:AvailabilityEvent` are both subclasses of `sv:PeriodRecord`, and `sv:startDate` and `sv:endDate` have `sv:PeriodRecord` as their domain.

Alternative : (a) Separate property pairs, such as `licenceStart`/`licenceEnd` and `availableFrom`/`availableUntil`. (b) Declare `sv:startDate` with domain `sv:Licence` and reuse it on availability events.

Reasoning : Option (b) is wrong in RDFS: a domain is an inference rule, so every availability event with a start date would be inferred to be a licence, which merges the two concepts the case asks to keep apart. Option (a) is safe but doubles the properties and every query that compares periods (Q8, Q10). With `sv:PeriodRecord` one pair of properties has one meaning, and the only thing RDFS infers is that licences and availability events are both period records (6 inferred type statements, Section 5.2).

6) Typed literals for dates, durations and numbers

Choice : Following the XSD datatypes [5], start, end and as-of dates are `xsd:date` (15 literals), runtimes are `xsd:duration` in ISO 8601 form such as `"PT1H40M"` (7), and episode numbers are `xsd:integer` (3). Each data property declares the matching `rdfs:range`.

Alternative : Plain strings such as “1 Jan 2026”, “1 h 40 min” or “E1”.

Reasoning : Typed dates can be compared directly in SPARQL with <= and >= (Q7, Q10 and Q11); as strings they would be compared alphabetically. A typed literal also makes a wrong value detectable: rdflib reports `"2026-02-30"` as an ill-typed `xsd:date` and `"1h 40m"` as an ill-typed `xsd:duration`, and all 25 literals in the submitted file are well formed. Episode numbers sort numerically, so Q4 returns E1, E2, E3 in order.

7) Availability status as a dated snapshot

Choice : `sv:availabilityStatus` points to a shared resource (`ex:Active` or `ex:Expired`), and `sv:asOfDate` records the day it was determined (2026-10-13). The start and end dates remain the source of truth; Q11 recomputes the status from them and compares.

Alternative : A plain string or boolean such as “active”, or no status at all.

Reasoning : The case asks for one expired and two active records, so the status has to be stated. A bare label silently goes stale: the “Active” on City Loop will be wrong after 2027-03-31. With `sv:asOfDate` the statement stays true (“active as of 2026-10-13”), Q11 finds any status that disagrees with the dates, and a shared `ex:Active` resource cannot be mistyped as “Actve”. Test M2 in Section 5.1 shows the check catching a wrong status.

4. Implementation

4.1 Building the Model

The vocabulary and the data were written directly in Turtle 1.1 [1] and saved as `streaming_catalogue.ttl`. **Part A** is the RDFS vocabulary [3]: 15 classes (with 5 `rdfs:subClassOf` statements) and 16 properties (11 object, 5 data), each with `rdfs:label` and `rdfs:comment`, and each property with `rdfs:domain` and `rdfs:range`. **Part B** is the data: 44 individuals, each with an `rdf:type` and an `rdfs:label`, linked through the object and data properties. Only standard Turtle, RDF, RDFS and XSD constructs are used, and the assumed values are marked with comments in the file.

Parsing, querying and RDFS reasoning were done in Python with rdflib 7.6.0 (Turtle parser and SPARQL 1.1 engine) and owlrl 7.6.2 (RDFS closure) [7], [8]. The script `run_checks.py` reproduces every number and query result in this report.

4.2 Loading and Querying

The Turtle file parsed without errors. The size of the graph was checked with a count query (Q1) :-

Graph	Triples
Vocabulary (Part A)	130

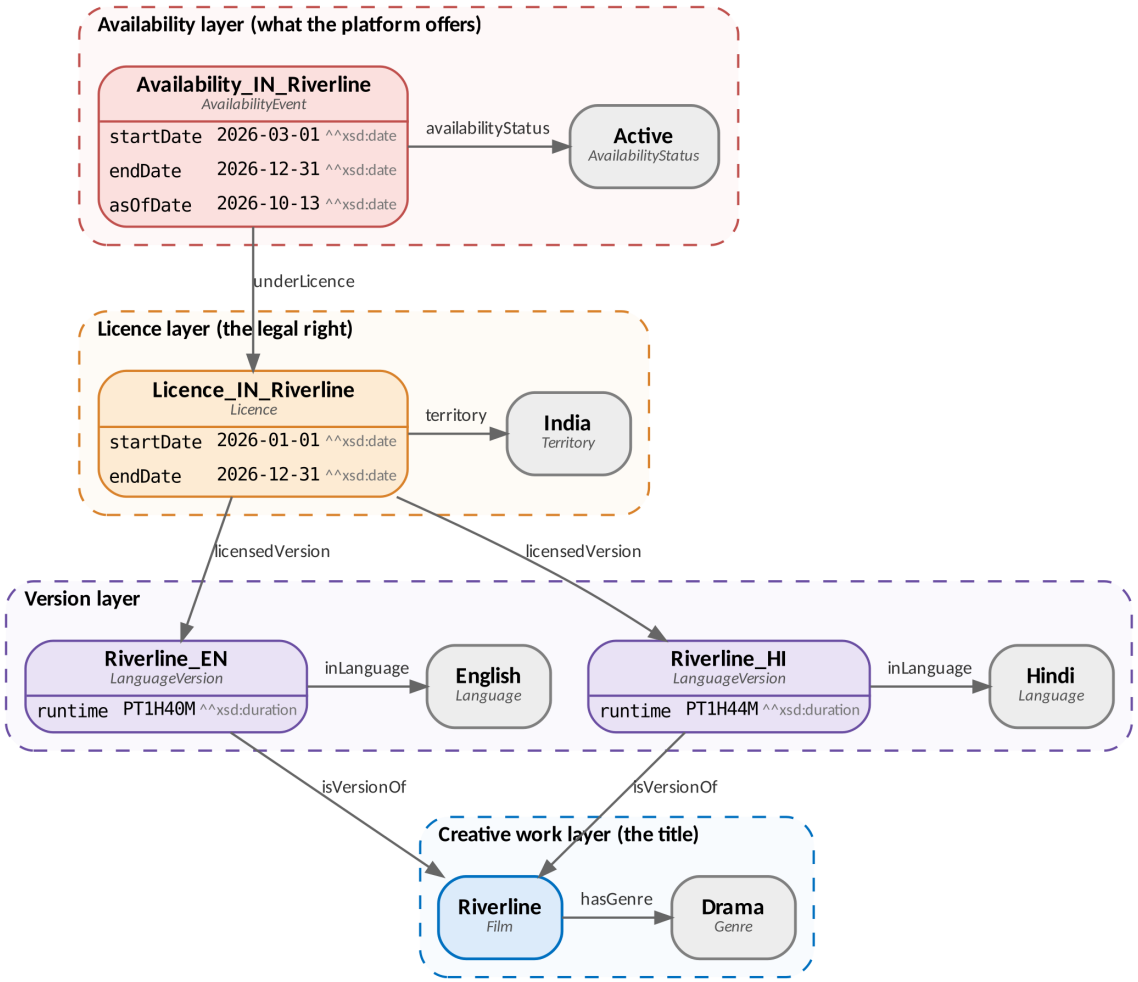
Graph	Triples
Data (Part B)	181
Explicit total (inference off)	**311**
Total with RDFS closure (inference on)	618
Inferred by RDFS	307

Of the 307 inferred triples, 12 are statements about this catalogue: 6 resources typed `sv:CreativeWork` and 6 typed `sv:PeriodRecord`. The other 295 are RDFS housekeeping: every resource is an `rdfs:Resource`, every class is a subclass of itself and of `rdfs:Resource`, every predicate used is an `rdf:Property`, and each literal is typed with its datatype. The closure total depends on the reasoner, so another RDFS tool will report a different number. The explicit count does not: 311 triples, of which 181 are data, well above the required 50.

All 12 SPARQL 1.1 queries [4] (Appendix B, also provided as `queries.rq`) were run twice: on the graph as written (inference off) and on its RDFS closure (inference on). Only Q1 (311 against 618 triples) and Q9 (0 against 6 rows) change. “Today” in Q7 is fixed at 2026-10-13, so the results can be repeated.

4.3 Visualizing a Subgraph

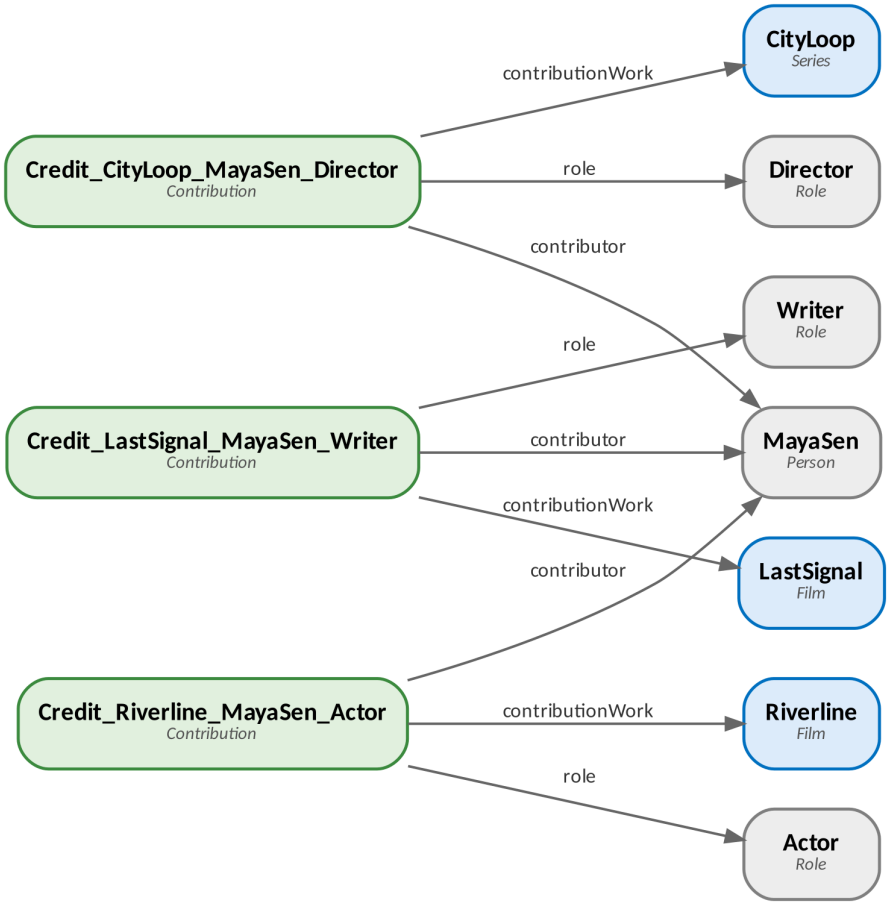
The subgraph around Riverline was drawn from the Turtle file with Graphviz, so every arrow is a triple in the file. Each box is a resource (the italic word is its `rdf:type`), and the rows inside a box are triples whose object is a typed literal.



Riverline Subgraph (Figure 2)
Availability event → licence → language versions → creative work. The four coloured bands are the four layers of Decision 4 and Decision 3.

- **Rights chain** : `Availability_IN_Riverline` ⟶ `underLicence` ⟶ `Licence_IN_Riverline` ⟶ `licensedVersion` ⟶ `Riverline_EN` and `Riverline_HI` ⟶ `isVersionOf` ⟶ `Riverline`. The territory (India) hangs off the licence, not off the film.
- **Typed literals** : the availability runs 2026-03-01 to 2026-12-31 with an as-of date of 2026-10-13, the licence runs 2026-01-01 to 2026-12-31 (all `xsd:date`), and the two versions run PT1H40M and PT1H44M (`xsd:duration`).
- **Separation** : in the creative work band, Riverline carries only its genre. No date, territory or status is attached to the film itself.

- **Reuse** : India, English, Hindi, Drama and Active are shared nodes. For example, Hindi is also the language of the three City Loop versions, and Active is also the status of `Availability_IN_CityLoop`.



Maya Sen's Credits (Figure 3)
Three `Contribution` nodes point to the same person (`MayaSen`), each with a different role and a different work (Decision 2).

5. Testing and Evidence

5.1 Validation

The model was checked in four ways :-

- **Syntax**: the Turtle file parses without errors (311 triples). All 25 typed literals (15 `xsd:date`, 7 `xsd:duration`, 3 `xsd:integer`) have a valid lexical form, and every `sv:` and `ex:` IRI used in the file is declared with an `rdf:type`.
- **Domain and range audit**: RDFS never reports a wrongly used property; it silently infers a type instead. So the RDFS closure was computed and, for each of the 44 individuals, the `sv:` classes it ended up with were compared with the classes expected from its declared type and the subclass hierarchy. There were 0 unexpected types, so no property is used outside its domain or range.

- **Data validation (Q10, Q11)**: every availability period starts before it ends and lies inside its licence period (Q10), and every stored status agrees with the dates on its as-of date (Q11). Both queries return 0 rows.
- **Separation (Q12)**: on the RDFS closure, no resource is both a creative work and a language version, licence, availability event or period record. The ASK query returns false.

To show that these checks are not empty, deliberate mistakes were made in in-memory copies of the graph. The submitted file was not changed.

Test	Deliberate mistake	Result
M1	End date of `Availability_IN_Riverline` moved to 2027-02-28, after its licence ends	Q10 returns 1 row
M2	`Availability_SG_LastSignal` marked `ex:Active`	Q11 returns 1 row (stored Active, computed Expired)
M3	A `sv:startDate` added to the film Riverline	Q12 returns true (Riverline is inferred to be a `sv:PeriodRecord`)
M4	The territory of `Licence_IN_Riverline` set to `ex:DevRao`	No check fires. Dev Rao is silently inferred to be a `sv:Territory` (Limitation 1)

5.2 Reasoning

To show that the RDFS vocabulary does real work, Q9 asks for every resource of type `sv:CreativeWork`. Nothing in the file is declared with that type directly.

Query	Inference off	Inference on
Q9: resources of type `sv:CreativeWork`	0 rows	6 rows: CityLoop, CityLoop_E1, CityLoop_E2, CityLoop_E3, LastSignal, Riverline
Q9 with `sv:PeriodRecord`	0 rows	6 rows: the 3 licences and the 3 availability events

The films are typed `sv:Film`, the series `sv:Series` and the episodes `sv:Episode`. Each of these is declared `rdfs:subClassOf sv:CreativeWork`, and the RDFS rule (x type C, C subClassOf D ⇒ x type D) lets a reasoner conclude that all six are creative works. In the same way `sv:Licence` and `sv:AvailabilityEvent` are subclasses of `sv:PeriodRecord`, which is also the domain of `sv:startDate` and `sv:endDate`.

5.3 Competency Questions

CQ	Query	Result	What it shows
CQ1: Who contributed to which work, and in what role?	Q2	9 rows. Dev Rao: Director of Riverline, Narrator of Last Signal, Writer of City Loop. Jo Paul: Writer, Director, Actor. Maya Sen: Actor, Writer, Director	The credit node lets one person hold different roles on different works (Decision 2)
CQ2: Which people hold more than one role?	Q3	Dev Rao, Jo Paul and Maya Sen: 3 roles on 3 works each	One IRI per person is reused across credits (Decisions 1, 2)

CQ	Query	Result	What it shows
CQ3: Which episodes belong to a series, in order, and how long is each?	Q4	City Loop E1 PT30M, E2 PT32M, E3 PT28M	`hasEpisode` and the typed `episodeNumber` give the order; runtimes sit on the versions (Decisions 3, 6)
CQ4: Which language versions does each title have?	Q5	7 rows: Riverline English and Hindi; Last Signal English and Malayalam; City Loop E1–E3 Hindi	Languages are shared resources; runtimes differ per version (Decision 3)
CQ5: Which licence covers which version, where and when?	Q6	7 rows. India: Riverline EN/HI 2026-01-01 to 2026-12-31; City Loop E1–E3 2026-07-01 to 2027-06-30. Singapore: Last Signal EN/ML 2026-01-01 to 2026-06-30	Licences are separate from works and list the versions they cover (Decisions 3, 4)
CQ6: Which titles are available on 2026-10-13, and where?	Q7	Riverline (India) and City Loop E1–E3 (India). Last Signal is not listed	Availability is computed by comparing `xsd:date` values (Decisions 4, 6)
CQ7: How does each licence period differ from its availability period?	Q8	India: licence 2026-01-01 to 2026-12-31, available from 2026-03-01. Singapore: available to 2026-06-15, licence to 2026-06-30. India: available 2026-08-01 to 2027-03-31, licence 2026-07-01 to 2027-06-30	Licence and availability are different things with different dates (Decision 4)
CQ8: Which resources are creative works?	Q9	0 rows without inference, 6 with it	RDFS subclass inference (Section 5.2)

The query behind CQ6 (Q7), which computes availability from the dates and does not rely on the stored status :-

```
SELECT DISTINCT ?title ?series ?territory ?from ?until
WHERE {
  BIND ("2026-10-13"^^xsd:date AS ?today)
  ?availability a sv:AvailabilityEvent ;
                sv:underLicence ?licence ;
                sv:startDate ?from ;
                sv:endDate   ?until .
  FILTER (?from <= ?today && ?today <= ?until)
  ?licence sv:licensedVersion ?v ;
           sv:territory/rdfs:label ?territory .
  ?v sv:isVersionOf ?work .
  ?work rdfs:label ?title .
  OPTIONAL { ?s sv:hasEpisode ?work . ?s rdfs:label ?series }
}
ORDER BY ?territory ?title
```

CQ6 Output (Q7)

title	series	territory	from	until
City Loop E1	City Loop	India	2026-08-01	2027-03-31
City Loop E2	City Loop	India	2026-08-01	2027-03-31
City Loop E3	City Loop	India	2026-08-01	2027-03-31
Riverline	–	India	2026-03-01	2026-12-31

CQ1 Output (Q2)

person	work	role
Dev Rao	City Loop	Writer

person	work	role
Dev Rao	Last Signal	Narrator
Dev Rao	Riverline	Director
Jo Paul	City Loop	Actor
Jo Paul	Last Signal	Director
Jo Paul	Riverline	Writer
Maya Sen	City Loop	Director
Maya Sen	Last Signal	Writer
Maya Sen	Riverline	Actor

CQ7 Output (Q8)

territory	licenceStart	availableFrom	availableUntil	licenceEnd	status
India	2026-01-01	2026-03-01	2026-12-31	2026-12-31	Active
Singapore	2026-01-01	2026-02-01	2026-06-15	2026-06-30	Expired
India	2026-07-01	2026-08-01	2027-03-31	2027-06-30	Active

6. Limitations and Reflection

6.1 Limitations

1) RDFS describes but does not enforce. `rdfs:domain` and `rdfs:range` are inference rules, not constraints. In test M4 a licence was linked to a person through `sv:territory`, and RDFS simply concluded that Dev Rao is a Territory; no error was raised. RDFS also cannot stop a licence with two territories, a missing end date or an end before a start. Rules like these need a constraint language such as SHACL (`sh:class`, `sh:maxCount`) [6]. Here they are approximated by the SPARQL checks Q10 to Q12.

2) Open World Assumption. A missing fact means "unknown", not "false". No episode has its own credits, but the graph cannot say that there are none, only that none are recorded. In the same way, Last Signal has no availability in India only because none is recorded.

3) Status is a snapshot. The "Active" on City Loop will be wrong after 2027-03-31 unless someone updates it. `sv:asOfDate` records when it was true and Q7 and Q11 recompute from the dates, but RDFS cannot compute a status by itself.

4) No inverse or derived properties in RDFS. `sv:hasEpisode` is stored in one direction only (the episode-to-series direction is reached with the path `^sv:hasEpisode`). Genre is not passed from a series to its episodes, and the series runtime is not stored, because a hand-typed total could disagree with its episodes. It would have to be summed with a query.

5) Rights are per version, not per series. The City Loop licence lists each episode version by hand. If an episode is added, nothing in RDFS adds it to the licence.

6) Coarse dates and territories. `xsd:date` has no time of day or time zone, and "end date is inclusive" is a documented convention that a machine cannot check. Territories are a flat list with no hierarchy (such as "South Asia" or "worldwide") and no exclusions.

6.2 Reflection

Anything that has its own dates, parties or status should be a node, not a property. Credits, language versions, licences and availability events all began as simple links (a person "directed" a film, a title "was in" India), and each one failed on a single sentence of the case: different roles, language versions, a licence that is not the same as availability. Modelling them as resources added hops to the queries (Q7 follows four links from an availability event to a title), but it meant that every question in Section 5.3 could be answered from the data without changing the vocabulary.

The inference switch (0 rows against 6 for creative works) also showed what RDFS adds. It checks nothing, but it lets a single query on `sv:CreativeWork` cover films, series and episodes. The mutation tests showed the other side: RDFS accepts wrong data silently, so the validation queries matter.

Next steps would be SHACL shapes for Q10 to Q12, credits for individual episodes, a territory hierarchy, and alignment with an existing vocabulary such as Schema.org (`schema:Movie`, `schema:TVSeries`, `schema:TVEpisode`) [11] or the BBC Programmes Ontology [9] for interoperability.

7. References and Acknowledgement

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[13] IETF. *RFC 2606: Reserved Top Level DNS Names.* 1999. https://www.rfc-editor.org/rfc/rfc2606

The Turtle file was parsed and queried with **rdflib 7.6.0**, and its RDFS closure was computed with **owlrl 7.6.2**. The diagrams were drawn from the Turtle file with **Graphviz 2.43.0**.

Claude (Anthropic) was used to restructure my earlier draft of this case into this report, to improve the model (language versions, episode numbers, a credit for every person on every title, separate namespaces, a status snapshot), to write the SPARQL queries and checking scripts, and to draft Sections 3 to 6. The scenario, my original draft and the choice of sample values are my own. Every number and query result in this report was produced by running the submitted files, and I have reviewed the final model and can explain and modify every part of it.

Appendix A. Complete Turtle File (streaming_catalogue.ttl)

The full submission graph: 311 triples (130 vocabulary and 181 data).

```
# ================================================================
# Streaming Catalogue and Rights (Case 2C) - CSE 4007 Semantic Web
# Author : Aayush Sharda (Reg. No. 235890136)
# File   : streaming catalogue.ttl   (Turtle 1.1, RDFS only)
#
# PART A  RDFS vocabulary   - namespace sv:
# PART B  Sample data       - namespace ex:
#
# All dates, runtimes, role and genre assignments are illustrative.
# Availability status is a snapshot taken on 2026-10-13 (asOfDate).
# Start and end dates are inclusive.
# ================================================================

@prefix sv:   <https://example.org/streaming/vocab#> .
@prefix ex:   <https://example.org/streaming/id/> .
@prefix rdf:  <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix xsd:  <http://www.w3.org/2001/XMLSchema#> .

# ================================================================
# PART A - RDFS VOCABULARY
# ================================================================

# ----------------------------------------------------------------
# A1. Classes
# ----------------------------------------------------------------

sv:CreativeWork a rdfs:Class ;
    rdfs:label "Creative work"@en ;
    rdfs:comment "A title in the catalogue: a film, a series or an episode. It carries no rights
  information."@en .

sv:Film a rdfs:Class ;
    rdfs:subClassOf sv:CreativeWork ;
    rdfs:label "Film"@en ;
    rdfs:comment "A standalone feature-length creative work."@en .

sv:Series a rdfs:Class ;
    rdfs:subClassOf sv:CreativeWork ;
    rdfs:label "Series"@en ;
    rdfs:comment "A creative work that is made up of episodes."@en .

sv:Episode a rdfs:Class ;
    rdfs:subClassOf sv:CreativeWork ;
    rdfs:label "Episode"@en ;
    rdfs:comment "One instalment of a series."@en .

sv:Person a rdfs:Class ;
    rdfs:label "Person"@en ;
    rdfs:comment "A contributor who can be credited on creative works."@en .

sv:Role a rdfs:Class ;
    rdfs:label "Role"@en ;
    rdfs:comment "A kind of contribution, such as director, actor, writer or narrator."@en .

sv:Genre a rdfs:Class ;
    rdfs:label "Genre"@en ;
    rdfs:comment "A category of creative work, such as Drama or Documentary."@en .

sv:Language a rdfs:Class ;
    rdfs:label "Language"@en ;
    rdfs:comment "A language in which a version of a creative work is offered."@en .

sv:Territory a rdfs:Class ;
    rdfs:label "Territory"@en ;
    rdfs:comment "A country or region in which a licence applies."@en .

sv:LanguageVersion a rdfs:Class ;
    rdfs:label "Language version"@en ;
    rdfs:comment "A creative work as offered in one language. It is the unit that has a runtime
  and that is licensed."@en .

sv:Contribution a rdfs:Class ;
```

```
    rdfs:label "Contribution"@en ;
    rdfs:comment "A credit that links one person, one role and one creative work."@en .

sv:PeriodRecord a rdfs:Class ;
    rdfs:label "Period record"@en ;
    rdfs:comment "A record that holds for a period with a start date and an end date (both
  inclusive)."@en .

sv:Licence a rdfs:Class ;
    rdfs:subClassOf sv:PeriodRecord ;
    rdfs:label "Licence"@en ;
    rdfs:comment "The legal right to stream specific language versions in one territory for a
  rights period."@en .

sv:AvailabilityEvent a rdfs:Class ;
    rdfs:subClassOf sv:PeriodRecord ;
    rdfs:label "Availability event"@en ;
    rdfs:comment "A period during which licensed versions are actually offered on the platform,
  under one licence."@en .

sv:AvailabilityStatus a rdfs:Class ;
    rdfs:label "Availability status"@en ;
    rdfs:comment "Whether an availability event is active or expired on its as-of date."@en .

# ------------------------------------------------------------------------
# A2. Object properties (resource -> resource)
# ------------------------------------------------------------------------

sv:hasGenre a rdf:Property ;
    rdfs:label "has genre"@en ;
    rdfs:comment "Links a creative work to its genre."@en ;
    rdfs:domain sv:CreativeWork ;
    rdfs:range sv:Genre .

sv:hasEpisode a rdf:Property ;
    rdfs:label "has episode"@en ;
    rdfs:comment "Links a series to one of its episodes."@en ;
    rdfs:domain sv:Series ;
    rdfs:range sv:Episode .

sv:isVersionOf a rdf:Property ;
    rdfs:label "is version of"@en ;
    rdfs:comment "Links a language version to the creative work it is a version of."@en ;
    rdfs:domain sv:LanguageVersion ;
    rdfs:range sv:CreativeWork .

sv:inLanguage a rdf:Property ;
    rdfs:label "in language"@en ;
    rdfs:comment "Links a language version to the language it is offered in."@en ;
    rdfs:domain sv:LanguageVersion ;
    rdfs:range sv:Language .

sv:contributionWork a rdf:Property ;
    rdfs:label "contribution work"@en ;
    rdfs:comment "The creative work a contribution (credit) belongs to."@en ;
    rdfs:domain sv:Contribution ;
    rdfs:range sv:CreativeWork .

sv:contributor a rdf:Property ;
    rdfs:label "contributor"@en ;
    rdfs:comment "The person named in a contribution (credit)."@en ;
    rdfs:domain sv:Contribution ;
    rdfs:range sv:Person .

sv:role a rdf:Property ;
    rdfs:label "role"@en ;
    rdfs:comment "The role the person holds in a contribution (credit)."@en ;
    rdfs:domain sv:Contribution ;
    rdfs:range sv:Role .

sv:licensedVersion a rdf:Property ;
    rdfs:label "licensed version"@en ;
    rdfs:comment "Links a licence to a language version that it covers."@en ;
    rdfs:domain sv:Licence ;
    rdfs:range sv:LanguageVersion .

sv:territory a rdf:Property ;
    rdfs:label "territory"@en ;
```

```
    rdfs:comment "The territory in which a licence applies."@en ;
    rdfs:domain sv:Licence ;
    rdfs:range sv:Territory .

sv:underLicence a rdf:Property ;
    rdfs:label "under licence"@en ;
    rdfs:comment "Links an availability event to the licence that allows it."@en ;
    rdfs:domain sv:AvailabilityEvent ;
    rdfs:range sv:Licence .

sv:availabilityStatus a rdf:Property ;
    rdfs:label "availability status"@en ;
    rdfs:comment "The status (active or expired) of an availability event on its as-of date."@en ;
    rdfs:domain sv:AvailabilityEvent ;
    rdfs:range sv:AvailabilityStatus .

# ------------------------------------------------------------------------
# A3. Data properties (resource -> typed literal)
# ------------------------------------------------------------------------

sv:episodeNumber a rdf:Property ;
    rdfs:label "episode number"@en ;
    rdfs:comment "Position of an episode within its series (1 for E1, 2 for E2, ...)."@en ;
    rdfs:domain sv:Episode ;
    rdfs:range xsd:integer .

sv:runtime a rdf:Property ;
    rdfs:label "runtime"@en ;
    rdfs:comment "Running time of a language version, written as an ISO 8601 duration such as
  PT1H40M."@en ;
    rdfs:domain sv:LanguageVersion ;
    rdfs:range xsd:duration .

sv:startDate a rdf:Property ;
    rdfs:label "start date"@en ;
    rdfs:comment "First day of a period record (inclusive)."@en ;
    rdfs:domain sv:PeriodRecord ;
    rdfs:range xsd:date .

sv:endDate a rdf:Property ;
    rdfs:label "end date"@en ;
    rdfs:comment "Last day of a period record (inclusive)."@en ;
    rdfs:domain sv:PeriodRecord ;
    rdfs:range xsd:date .

sv:asOfDate a rdf:Property ;
    rdfs:label "as-of date"@en ;
    rdfs:comment "The date on which the stated availability status was determined."@en ;
    rdfs:domain sv:AvailabilityEvent ;
    rdfs:range xsd:date .

# ========================================================================
# PART B - SAMPLE DATA
# ========================================================================

# ------------------------------------------------------------------------
# B1. Reference data: genres, languages, territories, roles, statuses
#     (reusable resources; each is created once and linked many times)
# ------------------------------------------------------------------------

ex:Drama a sv:Genre ;
    rdfs:label "Drama"@en .

ex:Documentary a sv:Genre ;
    rdfs:label "Documentary"@en .

ex:English a sv:Language ;
    rdfs:label "English"@en ;
    rdfs:seeAlso <http://dbpedia.org/resource/English language> .

ex:Hindi a sv:Language ;
    rdfs:label "Hindi"@en ;
    rdfs:seeAlso <http://dbpedia.org/resource/Hindi> .

ex:Malayalam a sv:Language ;
    rdfs:label "Malayalam"@en ;
    rdfs:seeAlso <http://dbpedia.org/resource/Malayalam> .
```

```
ex:India a sv:Territory ;
    rdfs:label "India"@en ;
    rdfs:seeAlso <http://dbpedia.org/resource/India> .

ex:Singapore a sv:Territory ;
    rdfs:label "Singapore"@en ;
    rdfs:seeAlso <http://dbpedia.org/resource/Singapore> .

ex:Director a sv:Role ;
    rdfs:label "Director"@en .

ex:Actor a sv:Role ;
    rdfs:label "Actor"@en .

ex:Writer a sv:Role ;
    rdfs:label "Writer"@en .

ex:Narrator a sv:Role ;
    rdfs:label "Narrator"@en .

ex:Active a sv:AvailabilityStatus ;
    rdfs:label "Active"@en .

ex:Expired a sv:AvailabilityStatus ;
    rdfs:label "Expired"@en .

# ------------------------------------------------------------------------
# B2. Contributors
# ------------------------------------------------------------------------

ex:DevRao a sv:Person ;
    rdfs:label "Dev Rao"@en .

ex:MayaSen a sv:Person ;
    rdfs:label "Maya Sen"@en .

ex:JoPaul a sv:Person ;
    rdfs:label "Jo Paul"@en .

# ------------------------------------------------------------------------
# B3. Creative works (titles only - no licence or availability data here)
#     Genre assignments are illustrative.
# ------------------------------------------------------------------------

ex:Riverline a sv:Film ;
    rdfs:label "Riverline"@en ;
    sv:hasGenre ex:Drama .

ex:LastSignal a sv:Film ;
    rdfs:label "Last Signal"@en ;
    sv:hasGenre ex:Documentary .

ex:CityLoop a sv:Series ;
    rdfs:label "City Loop"@en ;
    sv:hasGenre ex:Drama ;
    sv:hasEpisode ex:CityLoop E1, ex:CityLoop E2, ex:CityLoop E3 .

ex:CityLoop E1 a sv:Episode ;
    rdfs:label "City Loop E1"@en ;
    sv:episodeNumber "1"^^xsd:integer .

ex:CityLoop E2 a sv:Episode ;
    rdfs:label "City Loop E2"@en ;
    sv:episodeNumber "2"^^xsd:integer .

ex:CityLoop E3 a sv:Episode ;
    rdfs:label "City Loop E3"@en ;
    sv:episodeNumber "3"^^xsd:integer .

# ------------------------------------------------------------------------
# B4. Contributions (credits): one node per person + role + work.
#     Each person holds three different roles across the three titles.
#     Role choices are illustrative.
# ------------------------------------------------------------------------

ex:Credit Riverline DevRao Director a sv:Contribution ;
    rdfs:label "Riverline - Dev Rao - Director"@en ;
    sv:contributionWork ex:Riverline ;
```

```
    sv:contributor ex:DevRao ;
    sv:role ex:Director .

ex:Credit Riverline MayaSen Actor a sv:Contribution ;
    rdfs:label "Riverline - Maya Sen - Actor"@en ;
    sv:contributionWork ex:Riverline ;
    sv:contributor ex:MayaSen ;
    sv:role ex:Actor .

ex:Credit Riverline JoPaul Writer a sv:Contribution ;
    rdfs:label "Riverline - Jo Paul - Writer"@en ;
    sv:contributionWork ex:Riverline ;
    sv:contributor ex:JoPaul ;
    sv:role ex:Writer .

ex:Credit LastSignal JoPaul Director a sv:Contribution ;
    rdfs:label "Last Signal - Jo Paul - Director"@en ;
    sv:contributionWork ex:LastSignal ;
    sv:contributor ex:JoPaul ;
    sv:role ex:Director .

ex:Credit LastSignal DevRao Narrator a sv:Contribution ;
    rdfs:label "Last Signal - Dev Rao - Narrator"@en ;
    sv:contributionWork ex:LastSignal ;
    sv:contributor ex:DevRao ;
    sv:role ex:Narrator .

ex:Credit LastSignal MayaSen Writer a sv:Contribution ;
    rdfs:label "Last Signal - Maya Sen - Writer"@en ;
    sv:contributionWork ex:LastSignal ;
    sv:contributor ex:MayaSen ;
    sv:role ex:Writer .

ex:Credit CityLoop MayaSen Director a sv:Contribution ;
    rdfs:label "City Loop - Maya Sen - Director"@en ;
    sv:contributionWork ex:CityLoop ;
    sv:contributor ex:MayaSen ;
    sv:role ex:Director .

ex:Credit CityLoop JoPaul Actor a sv:Contribution ;
    rdfs:label "City Loop - Jo Paul - Actor"@en ;
    sv:contributionWork ex:CityLoop ;
    sv:contributor ex:JoPaul ;
    sv:role ex:Actor .

ex:Credit CityLoop DevRao Writer a sv:Contribution ;
    rdfs:label "City Loop - Dev Rao - Writer"@en ;
    sv:contributionWork ex:CityLoop ;
    sv:contributor ex:DevRao ;
    sv:role ex:Writer .

# ----------------------------------------------------------------------
# B5. Language versions. The runtime belongs to the version, because
#     two versions of one work can run for different lengths.
#     Version assignments and runtimes are illustrative.
# ----------------------------------------------------------------------

ex:Riverline EN a sv:LanguageVersion ;
    rdfs:label "Riverline (English)"@en ;
    sv:isVersionOf ex:Riverline ;
    sv:inLanguage ex:English ;
    sv:runtime "PT1H40M"^^xsd:duration .

ex:Riverline HI a sv:LanguageVersion ;
    rdfs:label "Riverline (Hindi)"@en ;
    sv:isVersionOf ex:Riverline ;
    sv:inLanguage ex:Hindi ;
    sv:runtime "PT1H44M"^^xsd:duration .

ex:LastSignal EN a sv:LanguageVersion ;
    rdfs:label "Last Signal (English)"@en ;
    sv:isVersionOf ex:LastSignal ;
    sv:inLanguage ex:English ;
    sv:runtime "PT1H18M"^^xsd:duration .

ex:LastSignal ML a sv:LanguageVersion ;
    rdfs:label "Last Signal (Malayalam)"@en ;
    sv:isVersionOf ex:LastSignal ;
```

```
    sv:inLanguage ex:Malayalam ;
    sv:runtime "PT1H20M"^^xsd:duration .

ex:CityLoop E1 HI a sv:LanguageVersion ;
    rdfs:label "City Loop E1 (Hindi)"@en ;
    sv:isVersionOf ex:CityLoop E1 ;
    sv:inLanguage ex:Hindi ;
    sv:runtime "PT30M"^^xsd:duration .

ex:CityLoop E2 HI a sv:LanguageVersion ;
    rdfs:label "City Loop E2 (Hindi)"@en ;
    sv:isVersionOf ex:CityLoop E2 ;
    sv:inLanguage ex:Hindi ;
    sv:runtime "PT32M"^^xsd:duration .

ex:CityLoop E3 HI a sv:LanguageVersion ;
    rdfs:label "City Loop E3 (Hindi)"@en ;
    sv:isVersionOf ex:CityLoop E3 ;
    sv:inLanguage ex:Hindi ;
    sv:runtime "PT28M"^^xsd:duration .

# ------------------------------------------------------------------------
# B6. Regional licences (the legal right: versions + territory + period)
#     Licence assignments and dates are illustrative.
# ------------------------------------------------------------------------

ex:Licence IN Riverline a sv:Licence ;
    rdfs:label "India licence - Riverline"@en ;
    sv:licensedVersion ex:Riverline_EN, ex:Riverline_HI ;
    sv:territory ex:India ;
    sv:startDate "2026-01-01"^^xsd:date ;
    sv:endDate "2026-12-31"^^xsd:date .

ex:Licence SG LastSignal a sv:Licence ;
    rdfs:label "Singapore licence - Last Signal"@en ;
    sv:licensedVersion ex:LastSignal EN, ex:LastSignal ML ;
    sv:territory ex:Singapore ;
    sv:startDate "2026-01-01"^^xsd:date ;
    sv:endDate "2026-06-30"^^xsd:date .

ex:Licence_IN_CityLoop a sv:Licence ;
    rdfs:label "India licence - City Loop episodes"@en ;
    sv:licensedVersion ex:CityLoop_E1_HI, ex:CityLoop_E2_HI, ex:CityLoop_E3_HI ;
    sv:territory ex:India ;
    sv:startDate "2026-07-01"^^xsd:date ;
    sv:endDate "2027-06-30"^^xsd:date .

# ------------------------------------------------------------------------
# B7. Availability events (when the platform actually offers the titles)
#     Snapshot date 2026-10-13, end dates inclusive:
#       one expired record and two active records.
# ------------------------------------------------------------------------

ex:Availability IN Riverline a sv:AvailabilityEvent ;
    rdfs:label "Availability - Riverline in India"@en ;
    sv:underLicence ex:Licence IN Riverline ;
    sv:startDate "2026-03-01"^^xsd:date ;
    sv:endDate "2026-12-31"^^xsd:date ;
    sv:availabilityStatus ex:Active ;
    sv:asOfDate "2026-10-13"^^xsd:date .

ex:Availability SG LastSignal a sv:AvailabilityEvent ;
    rdfs:label "Availability - Last Signal in Singapore"@en ;
    sv:underLicence ex:Licence SG LastSignal ;
    sv:startDate "2026-02-01"^^xsd:date ;
    sv:endDate "2026-06-15"^^xsd:date ;
    sv:availabilityStatus ex:Expired ;
    sv:asOfDate "2026-10-13"^^xsd:date .

ex:Availability IN CityLoop a sv:AvailabilityEvent ;
    rdfs:label "Availability - City Loop in India"@en ;
    sv:underLicence ex:Licence IN CityLoop ;
    sv:startDate "2026-08-01"^^xsd:date ;
    sv:endDate "2027-03-31"^^xsd:date ;
    sv:availabilityStatus ex:Active ;
    sv:asOfDate "2026-10-13"^^xsd:date .
```

Appendix B. SPARQL Queries (queries.rq)

Every query starts with the same four prefixes, shown once here. In `queries.rq` each query is complete on its own.

```
PREFIX sv:   <https://example.org/streaming/vocab#>
PREFIX ex:   <https://example.org/streaming/id/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd:  <http://www.w3.org/2001/XMLSchema#>
```

Q1. Size of the graph

```
SELECT (COUNT(*) AS ?triples)
WHERE { ?s ?p ?o }
```

Q2. CQ1 - Who contributed to which work, and in what role?

```
SELECT ?person ?work ?role
WHERE {
  ?credit a sv:Contribution ;
          sv:contributor/rdfs:label   ?person ;
          sv:contributionWork/rdfs:label ?work ;
          sv:role/rdfs:label          ?role .
}
ORDER BY ?person ?work
```

Q3. CQ2 - Which people hold more than one role, and on how many works?

```
SELECT ?person (COUNT(DISTINCT ?role) AS ?roles) (COUNT(DISTINCT ?work) AS ?works)
WHERE {
  ?credit a sv:Contribution ;
          sv:contributor ?p ;
          sv:contributionWork ?work ;
          sv:role ?role .
  ?p rdfs:label ?person .
}
GROUP BY ?person
HAVING (COUNT(DISTINCT ?role) > 1)
ORDER BY ?person
```

Q4. CQ3 - Which episodes belong to a series, in order, and how long is each one?

```
SELECT ?series ?number ?episode ?language ?runtime
WHERE {
  ?s sv:hasEpisode ?e ;
     rdfs:label ?series .
  ?e sv:episodeNumber ?number ;
     rdfs:label ?episode .
  ?v sv:isVersionOf ?e ;
     sv:inLanguage/rdfs:label ?language ;
     sv:runtime ?runtime .
}
ORDER BY ?series ?number
```

Q5. CQ4 - Which language versions does each work have, and how long does each version run?

```
SELECT ?work ?language ?runtime
WHERE {
  ?v a sv:LanguageVersion ;
     sv:isVersionOf/rdfs:label ?work ;
     sv:inLanguage/rdfs:label  ?language ;
     sv:runtime                ?runtime .
}
ORDER BY ?work ?language
```

Q6. CQ5 - Which licence covers which version, in which territory and for which rights period?

```
SELECT ?territory ?version ?start ?end
WHERE {
  ?licence a sv:Licence ;
           sv:licensedVersion ?v ;
           sv:territory/rdfs:label ?territory ;
           sv:startDate ?start ;
           sv:endDate   ?end .
  ?v rdfs:label ?version .
}
ORDER BY ?territory ?start ?version
```

Q7. CQ6 - Which titles are available on the reference date, and where? (computed from the dates, not from the stored status)

```
SELECT DISTINCT ?title ?series ?territory ?from ?until
WHERE {
  BIND ("2026-10-13"^^xsd:date AS ?today)
  ?availability a sv:AvailabilityEvent ;
                sv:underLicence ?licence ;
                sv:startDate ?from ;
                sv:endDate   ?until .
  FILTER (?from <= ?today && ?today <= ?until)
  ?licence sv:licensedVersion ?v ;
           sv:territory/rdfs:label ?territory .
  ?v sv:isVersionOf ?work .
  ?work rdfs:label ?title .
  OPTIONAL { ?s sv:hasEpisode ?work . ?s rdfs:label ?series }
}
ORDER BY ?territory ?title
```

Q8. CQ7 - How does each licence period differ from the availability period that runs under it?

```
SELECT ?territory ?licenceStart ?availableFrom ?availableUntil ?licenceEnd ?status
WHERE {
  ?availability a sv:AvailabilityEvent ;
                sv:underLicence ?licence ;
                sv:startDate ?availableFrom ;
                sv:endDate   ?availableUntil ;
                sv:availabilityStatus/rdfs:label ?status .
  ?licence sv:territory/rdfs:label ?territory ;
           sv:startDate ?licenceStart ;
           sv:endDate   ?licenceEnd .
}
ORDER BY ?licenceStart
```

Q9. CQ8 - Which resources are creative works? (needs RDFS inference: nothing is asserted as sv:CreativeWork)

```
SELECT ?work
WHERE { ?work a sv:CreativeWork }
ORDER BY ?work
```

Q10. VALIDATION 1 - availability must start before it ends and lie inside its licence period.

```
SELECT ?availability ?from ?until ?licenceStart ?licenceEnd
WHERE {
  ?availability sv:underLicence ?licence ;
                sv:startDate ?from ;
                sv:endDate   ?until .
  ?licence sv:startDate ?licenceStart ;
           sv:endDate   ?licenceEnd .
  FILTER (?from > ?until || ?from < ?licenceStart || ?until > ?licenceEnd)
}
```

Q11. VALIDATION 2 - the stored status must agree with the dates on the as-of date.

```
SELECT ?availability ?stored ?computed
WHERE {
  ?availability a sv:AvailabilityEvent ;
                sv:startDate ?from ;
                sv:endDate   ?until ;
                sv:asOfDate  ?asOf ;
                sv:availabilityStatus/rdfs:label ?stored .
  BIND (IF(?from <= ?asOf && ?asOf <= ?until, "Active",
        IF(?asOf > ?until, "Expired", "Upcoming")) AS ?computed)
  FILTER (STR(?stored) != ?computed)
}
```

Q12. VALIDATION 3 - creative works must stay separate from versions, licences and availability events. Run WITH RDFS inference on.

```
ASK {
  ?x a sv:CreativeWork ;
     a ?other .
  VALUES ?other { sv:LanguageVersion sv:Licence sv:AvailabilityEvent sv:PeriodRecord }
}
```